LEARNING CONTROLLABLE TEMPO AND INTENSITY IN STOCHASTIC AUDIO USING NEURAL AUDIO CODECS: A STUDY ON THE BEANSHAKE DATASET

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ABSTRACT

This paper investigates whether a neural audio codec can learn interpretable and controllable parameters corresponding to perceptual dimensions of tempo and intensity in a non-musical, stochastic sound source. Using the Beanshake dataset, which contains recordings of an aluminum bottle being shaken with a varying number of beans inside, this work explores multiple feature extraction strategies to parameterize the perceptual attributes of tempo and intensity of the shaking. Specifically, this study evaluates amplitude envelope representations, frame-level RMS energy, and onset-derived tempo estimates expressed in beats per minute (BPM). These features are used to condition a recurrent neural network-based audio codec model with the goal of enabling user control over generated audio. Our results demonstrate that energy-based features were more successful in being a control mechanism for perceived intensity and tempo, while methods for onset detection fail to produce stable or perceptually meaningful control. These findings highlight fundamental challenges in transferring music-based temporal representations to stochastic audio signals and suggest the need for alternative formulations of temporal structure in non-periodic domains.

1. INTRODUCTION

Recent advances in neural audio codecs have enabled high-quality compression and reconstruction of audio signals while also opening new possibilities for controllable audio synthesis. Models using neural audio codecs learn compact latent representations that can be conditioned to influence generation. However, these latent spaces are typically difficult to interpret, and the identification of meaningful control parameters remains an open problem.

In music information retrieval (MIR), perceptual dimensions such as tempo and dynamics are well-defined and can be (somewhat) reliably extracted using signal processing approaches. In contrast, many real-world sounds, particularly those arising from physical interactions, exhibit stochastic and quasi-periodic behavior that does not conform to traditional musical structure. This raises the question of whether such perceptual dimensions can be meaningfully defined and learned in these contexts.

This work investigates whether a neural audio codec can learn disentangled and controllable representations of tempo and intensity from a simple but informative audio stimulus. The Beanshake dataset provides a controlled environment in which there is a relatively stable shaking pattern with variations in shaking behavior and object density producing complex acoustic patterns. To assess the effectiveness of intensity and temporal structure as conditioning signals for controllable audio generation, audio outputs are perceptually evaluated based on how closely they resemble the natural sound of someone shaking a container of beans while changing the perceptual parameters of tempo and intensity.

2. RELATED WORKS

Neural audio synthesis has advanced rapidly with the development of deep generative models capable of producing high-fidelity waveforms directly from learned representations. Early work by Van den Oord et al. introduced WaveNet, an autoregressive model that generates raw audio sample-by-sample using dilated convolutions [1]. This approach demonstrated that neural networks could model complex temporal dependencies in audio, though at the cost of high computational expense and limited controllability. Subsequent work has focused on improving efficiency and enabling conditional generation. Kumar et al. proposed MelGAN, a generative adversarial network (GAN)-based vocoder that significantly reduces inference time while maintaining perceptual quality [2]. By framing waveform synthesis as a conditional generation problem, MelGAN highlights the importance of auxiliary features in guiding audio generation, although these features are typically low-level representations such as mel-spectrograms rather than interpretable control parameters. In parallel, research on differentiable digital signal processing (DDSP) has demonstrated the effectiveness of conditioning neural synthesis models on interpretable features such as loudness and pitch. Engel et al. showed that explicitly modeling such features can improve both control and interpretability in generative systems [3]. This work inspires the use of

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engineered features, such as RMS energy and amplitude envelopes, as proxies for perceptual dimensions in non-musical audio. However, relatively little work has explored how these concepts transfer to stochastic sounds.

More broadly, system-level approaches to generative audio have emphasized the importance of modular and controllable design. Wyse et al. propose an integrated architecture for generative audio modeling that combines multiple synthesis and control components into a unified framework [4]. Their work highlights the role of structured parameterization and flexible pipelines in enabling user-driven audio generation, reinforcing the importance of designing meaningful control representations.

Prior work in MIR has established robust deep learning techniques for onset detection and tempo estimation. Böck and Schedl proposed methods for detecting note onsets in polyphonic music, which can be used to estimate inter-onset intervals and derive tempo [5]. These methods rely on assumptions of periodicity and rhythmic regularity that may not hold in stochastic audio signals. The present work builds on these foundations by investigating whether traditional MIR-derived features, specifically tempo and intensity, can be learned by feature-conditioned neural audio codecs when applied in a non-musical, stochastic audio domain.

3. METHODS

3.1 Beanshake Dataset

The Beanshake dataset consists of eleven audio recordings, each approximately forty seconds in duration, capturing the sound of an aluminum water bottle being shaken with varying numbers of garbanzo beans inside. The primary controlled variable in the dataset is the number of beans, referred to as **BeanCount**. BeanCount ranged from 1 to 1024 beans, increasing logarithmically. Changes in BeanCount produce noticeable differences in timbre, with lower counts yielding sparse, metallic impacts and higher counts producing denser, more noise-like textures. Although the dataset contains stochastic sounds, there is a degree of regularity in the shaking rate (visible in **Figure 1**). This motivated the investigation of whether such temporal structure could be meaningfully extracted and learned.

3.2 Feature Extraction

To parameterize intensity and temporal structure, three feature extraction methods were explored. The first method computes a smoothed amplitude envelope. To extract the amplitude envelope, each audio signal is converted to mono, rectified using a full-wave operation, and passed through a low-pass filter to suppress high-frequency fluctuations. The degree of smoothing is controlled by a parameter specifying the window size in milliseconds. The resulting envelope is then downsampled to a fixed frame rate of 75 frames per second. This representation is intended to capture macro-level variations in energy and rhythmic structure. The second method extracts two features: RMS energy in decibels and an estimate of shake rate expressed in BPM. RMS energy is computed on a frame-by-frame basis using the librosa library [6] and converted to a decibel scale, with a floor applied to silence regions. To estimate tempo, onset detection is performed to identify transient events, from which inter-onset intervals (IOIs) are calculated. These IOIs are interpolated across the full time axis and converted to BPM using the standard relationship between period and frequency. The third method refines the tempo estimation process by deriving BPM from peaks in the amplitude envelope rather than directly from onset detection. Peaks are identified using prominence and minimum distance constraints, and the resulting IOIs are interpolated and converted to BPM. This approach aims to produce a more stable estimate of temporal structure by leveraging the smoothed envelope representation. An additional feature set incorporating multiple descriptors, including spectral flux and onset strength, was considered but ultimately rejected due to certain features being highly correlated with each other, therefore rendering them redundant and introducing unnecessary complexity in the control space.

3.3 Model and Experimental Setup

A recurrent neural network-based audio codec model similar to that of [4] was trained on the extracted acoustic features as conditioning inputs. The central research question guiding this experiment is whether the model can learn disentangled and controllable representations of tempo and intensity in a stochastic audio signal. The model was trained across multiple configurations, varying both the feature representations and training duration. Performance was evaluated qualitatively by assessing the perceptual effects of manipulating the conditioning parameters during audio generation.

4. RESULTS

Analysis of the original audio recordings reveals clear timbral variation as a function of BeanCount, with higher counts producing more continuous and noise-like textures. This suggests that the dataset contains meaningful acoustic variation that could potentially be captured by learned representations.

The amplitude envelope method demonstrates the strongest success in controlling perceived intensity and tempo. In particular, envelope values in the range of approximately 0.05 to 0.13 produce the most perceptually clear and stable shaking sounds which seem to increase and decrease in speed with the change in the parameter value. The choice of smoothing parameter proved to be critical through an ablation study visible in **Figure 2**. A window of 80 milliseconds provides a balance between preserving transient structure and reducing noise, whereas smaller windows result in overly noisy signals and larger windows oversmooth certain temporal dynamics which are more prominent in the audio evaluation.

The RMS and BPM feature combination yields mixed results. While RMS energy provides a reliable control over

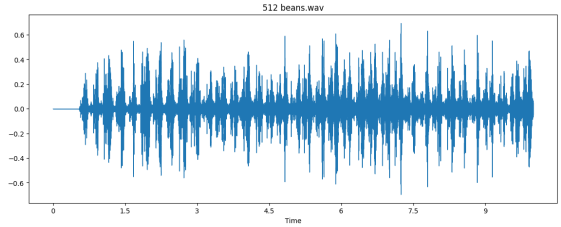

Figure 1: Original example audio waveform for 512 beans, demonstrating clear peaks for shake onset which are relatively evenly spaced

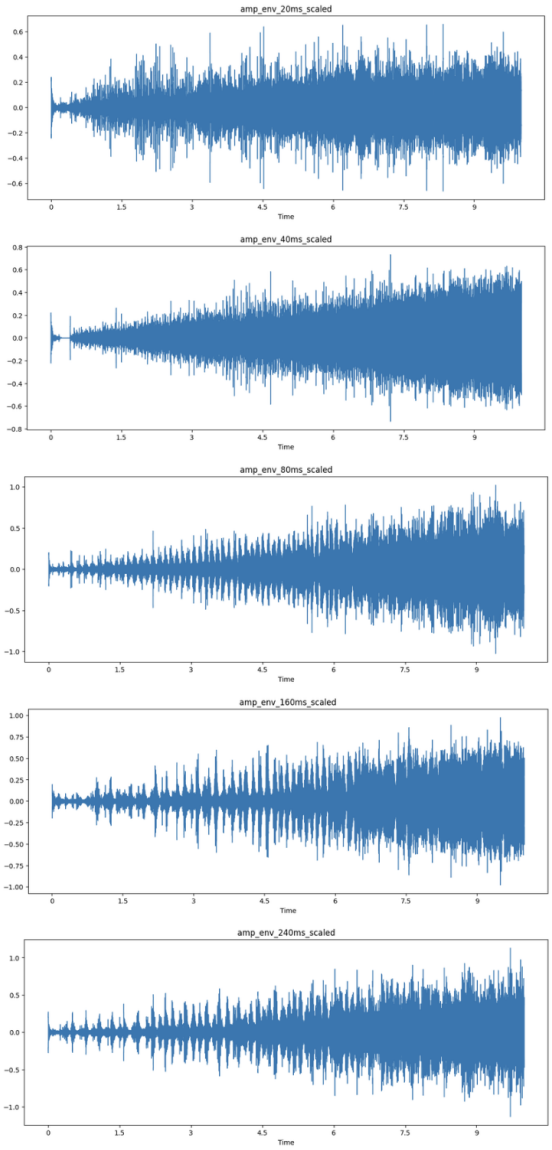

Figure 2: Audio waveforms of generated sounds from the neural audio codec trained on amplitude envelope while changing the smoothing window size. Throughout the generation inference, the amplitude_envelope parameter was linearly interpolated from 0 to 0.20.

perceived loudness, with silence occurring below approximately -40 dB and increasing amplitude above this threshold as seen in **Figure 3**, the BPM parameter does not pro-

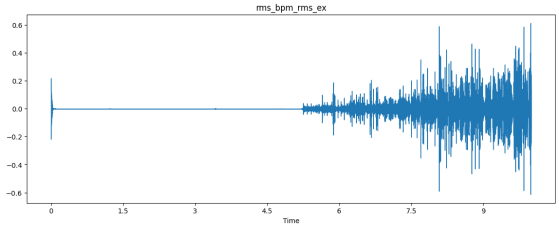

Figure 3: Audio waveforms of generated sounds from the neural audio codec trained on RMS in dB and shake rate in BPM. Throughout the audio, the RMS parameter was linearly interpolated from -80.0 to 0 dB while the shake rate in BPM was held constant at the midpoint of 750.

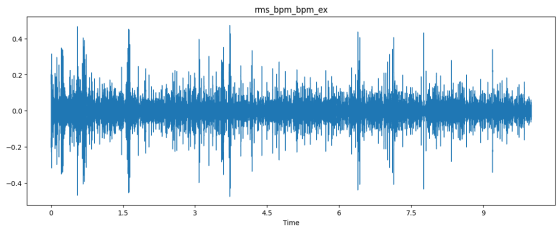

Figure 4: Audio waveforms of generated sounds from the neural audio codec trained on RMS in dB and shake rate in BPM. Throughout the audio, the BPM parameter was linearly interpolated from 0 to 1500 while the RMS was held constant at -20 dB.

duce meaningful perceptual changes visible in **Figure 4**. Increasing the number of training epochs from 15 to 75 does not improve the model's ability to respond to BPM variations and did not produce any meaningful difference in auditory perception nor in the waveform shape.

The improved BPM estimation method based on amplitude envelope peaks introduces some variation in the generated audio, particularly at higher BPM values where occasional clustered "rattle" events occur. However, these effects are inconsistent and do not correspond to a stable or controllable sense of tempo.

Audio examples with their respective waveforms and spectrograms and details on training configurations can be seen and explored in the classnn-pereiraa10 github repo.

5. DISCUSSION

The results indicate that intensity, as represented by energy-based features, can be learned and controlled by the model to a limited extent. In contrast, tempo proves difficult to parameterize and control in this domain. This difficulty arises from the lack of stable periodic structure in the underlying signal, as well as the presence of overlapping events that obscure clear temporal boundaries.

These findings suggest that traditional representations of tempo, such as BPM derived from inter-onset intervals, may not be appropriate for stochastic audio signals. Instead, temporal structure in such signals may be better characterized in terms of event density or probabilistic distributions of inter-event timing.

Furthermore, the strong dependence of results on feature design underscores the importance of selecting representations that align with the intrinsic properties of the signal. Small changes in preprocessing, such as the choice of smoothing window, can significantly impact the learnability of the features.

6. CONCLUSION AND FUTURE WORK

This study demonstrates that neural audio codecs can learn partially controllable representations of intensity in stochastic audio signals, but struggle to capture and expose tempo as a meaningful control parameter. The limitations observed in tempo modeling highlight a broader challenge in extending music-based representations to non-musical domains.

Future work should explore alternative representations of temporal structure, such as event density or learned latent variables, as well as approaches that incorporate physical modeling of sound generation processes. Expanding the dataset and introducing more controlled variations in shaking behavior may also improve the robustness of learned representations. Additionally, using manually annotated parameters, while tedious, could help evade any noise in training due to systematic error in any of the steps before BPM calculation, like during onset detection. Ultimately, developing effective control mechanisms for stochastic audio remains an open and promising area of research. While this work mostly focused on the features extracted to use as parameters to train the model, there are additional parameters of the model itself which could be explored to see how they impact the results, particularly exploring why limited training epochs were helpful and to determine whether the model is converging too early.

These results suggest that the challenge of controllable audio generation lies not only in model architecture, but also in how perceptual concepts are defined and represented. While intensity maps naturally onto measurable signal properties, tempo in stochastic systems lacks a clear and consistent definition. This suggests that future research should focus not only on improving models, but also on rethinking the conceptual frameworks used to describe and parameterize complex audio signals.

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